

Section: Particles and Waves

Topic: Spectra

A student observes the line spectrum of light from a hot gas.

(a)

State the type of spectrum observed.

✔ A line emission spectrum.

This consists of **bright lines** at specific wavelengths on a dark background, each corresponding to a photon emitted as electrons transition between energy levels.

(b)

Explain how this spectrum provides evidence for the existence of discrete energy levels in atoms.

✔ Each spectral line corresponds to a **photon of specific energy** being emitted.

Since the energy of the photon is:

$$E = hf$$

and only **specific frequencies** (or wavelengths) are emitted, this implies that electrons **only transition between fixed energy levels**.

The pattern of lines is unique to each element, indicating that energy levels are **quantised** — not continuous.

🧠 Revision Tips:

- **Emission spectra:** bright lines → electron **falls** to a lower energy level
- **Absorption spectra:** dark lines → electron **absorbs** energy to rise to a higher level
- Only certain transitions are allowed → energy levels are **discrete**